# 2 ALARM NUMBERS AND BRIEF DESCRIPTIONS

# 2.1 FOR SERIES 16*i*, 18*i*, 20*i*, 21*i*, AND Power Mate *i*

Servo alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 360 | | | | Pulse coder checksum error (built-in) | 3.3.7 (1) |
| 361 | | | | Pulse coder phase error (built-in) | 3.3.7 (1) |
| 363 | | | | Clock error (built-in) | 3.3.7 (1) |
| 364 | | | | Soft phase alarm (built-in) | 3.3.7 (1) |
| 365 | | | | LED error (built-in) | 3.3.7 (1) |
| 366 | | | | Pulse error (built-in) | 3.3.7 (1) |
| 367 | | | | Count error (built-in) | 3.3.7 (1) |
| 368 | | | | Serial data error (built-in) | 3.3.7 (3) |
| 369 | | | | Data transfer error (built-in) | 3.3.7 (3) |
| 380 | | | | LED error (separate) | 3.3.7 (2) |
| 381 | | | | Pulse coder phase error (separate) | 3.3.7 (2) |
| 382 | | | | Count error (separate) | 3.3.7 (2) |
| 383 | | | | Pulse error (separate) | 3.3.7 (2) |
| 384 | | | | Soft phase alarm (separate) | 3.3.7 (2) |
| 385 | | | | Serial data error (separate) | 3.3.7 (3) |
| 386 | | | | Data transfer error (separate) | 3.3.7 (3) |
| 417 | | | | Invalid parameter | 3.3.6 |
| 421 | | | | Excessive semi-full error | 3.3.8 |
| 430 | | | | Servomotor overheat | 3.3.5 |
| 431 | | | 03 | Converter: main circuit overload | 3.1 |
| 432 | | | 06 | Converter: control undervoltage/open phase | 3.1 |
| 433 | | | 04 | Converter: DC link undervoltage | 3.1 |
| 434 | 2 | | | Inverter: control power supply undervoltage | 3.2 |
| 435 | 5 | | | Inverter: DC link undervoltage | 3.2 |
| 436 | | | | Soft thermal (OVC) | 3.3.3 |
| 437 | | | 01 | Converter: input circuit overcurrent | 3.1 |
| 438 | 8<br>9<br>A<br>b<br>C<br>d<br>E | | | Inverter: current alarm (L axis)<br>(M axis)<br>(N axis)<br>(L and M axes)<br>(M and N axes)<br>(L and N axes)<br>(L, M, and N axes) | 3.2 |
| 439 | | | 07 | Converter: DC link overvoltage | 3.1 |
| 440 | | | 08 | Converter excessive deceleration power | 3.1 |

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 441 | | | | Current offset error | 3.3.8 |
| 442 | | | 05 | Converter: DC link charging/inverter DB | 3.1 |
| 443 | | | 02 | Converter: cooling fan stopped | 3.1 |
| 444 | 1 | | | Inverter: internal cooling fan stopped | 3.2 |
| 445 | | | | Soft disconnection alarm | 3.3.4 |
| 446 | | | | Hard disconnection alarm | Not issued |
| 447 | | | | Hard disconnection alarm (separate) | 3.3.4 |
| 448 | | | | Feedback mismatch alarm | 3.3.8 |
| 449 | 8.<br>9.<br>A.<br>b.<br>C.<br>d.<br>E. | | | Inverter: IPM alarm (L axis)<br>(M axis)<br>(N axis)<br>(L and M axes)<br>(M and N axes)<br>(L and N axes)<br>(L, M, and N axes) | 3.2 |
| 453 | | | | Soft disconnection alarm ($\alpha$ pulse coder) | 3.3.4 |

Spindle alarm (n represents a spindle number. Example: n = 1 for the first spindle)

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 749 | | A | | Program ROM error | 3.4 |
| 749 | | A0 | | Program ROM error | 3.4 |
| 749 | | A1 | | Program RAM error | 3.4 |
| 749 | | A2 | | Program RAM error | 3.4 |
| 749 | | A3 | | SPM control circuit clock error | 3.4 |
| 749 | | A4 | | SRAM parity error | 3.4 |
| 7n01 | | 01 | | Motor overheat | 3.4 |
| 7n02 | | 02 | | Excessive speed deviation | 3.4 |
| 7n03 | | 03 | | DC link fuse blown | 3.4 |
| 7n04 | | 04 | 06 | Input power supply open phase and power supply failure | 3.4 |
| 7n07 | | 07 | | Excessive speed | 3.4 |
| 7n09 | | 09 | | Main circuit overload | 3.4 |
| 7n11 | | 11 | 07 | DC link overvoltage | 3.4 |
| 7n12 | | 12 | | DC link overcurrent/IPM alarm | 3.4 |
| 750 | | 13 | | CPU internal data memory error | 3.4 |
| 7n15 | | 15 | | Output switching/spindle switching alarm | 3.4 |
| 750 | | 16 | | RAM error | 3.4 |
| 750 | | 19 | | Excessive offset of the phase U current detection circuit | 3.4 |
| 750 | | 20 | | Excessive offset of the phase V current detection circuit | 3.4 |

(n represents a spindle number. Example: n = 1 for the first spindle)

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 749 | | 24 | | Serial transfer data error | 3.4 |
| 749 | | 25 | | Setting data transfer stopped | 3.4 |
| 7n26 | | 26 | | Cs contour control speed detection signal disconnected | 3.4 |
| 7n27 | | 27 | | Position coder signal disconnected | 3.4 |
| 7n28 | | 28 | | Cs contour control position detection signal disconnected | 3.4 |
| 7n29 | | 29 | | Short-period overload | 3.4 |
| 7n30 | | 30 | 01 | PSM main circuit overcurrent | 3.4 |
| 7n31 | | 31 | | Speed detection signal disconnected and motor lock alarm | 3.4 |
| 750 | | 32 | | Serial communication LSI internal RAM error | 3.4 |
| 7n33 | | 33 | 05 | DC link precharge failure | 3.4 |
| 7n34 | | 34 | | Parameter data out of the specifiable range | 3.4 |
| 7n35 | | 35 | | Too large of a gear ratio is specified. | 3.4 |
| 7n36 | | 36 | | Error counter overflow | 3.4 |
| 7n37 | | 37 | | Speed detector parameter error | 3.4 |
| 7n39 | | 39 | | Cs contour control one-rotation signal detection error | 3.4 |
| 7n40 | | 40 | | Cs contour control one-rotation signal not detected | 3.4 |
| 7n41 | | 41 | | Position coder one-rotation signal detection error | 3.4 |
| 7n42 | | 42 | | Position coder one-rotation signal not detected | 3.4 |
| 7n43 | | 43 | | Differential speed mode position coder signal disconnected | 3.4 |
| 7n44 | | 44 | | A/D conversion error | 3.4 |
| 7n46 | | 46 | | Position coder one-rotation signal detection error during thread cutting | 3.4 |
| 7n47 | | 47 | | Position coder signal error | 3.4 |
| 7n48 | | 49 | | Excessive differential speed conversion result | 3.4 |
| 7n50 | | 50 | | Excessive speed command calculation value during spindle synchronization | 3.4 |
| 7n51 | | 51 | 04 | DC link undervoltage | 3.4 |
| 7n52 | | 52 | | ITP signal error I | 3.4 |
| 7n53 | | 53 | | ITP signal error II | 3.4 |
| 7n54 | | 54 | | Current overload alarm | 3.4 |
| 7n55 | | 55 | | Power line state error at spindle switching/output switching | 3.4 |
| 7n56 | | 56 | | Control circuit cooling fan stopped | 3.4 |
| 7n56 | | 57 | 08 | Excessive regenerated power | 3.4 |
| 7n58 | | 58 | 03 | PSM main circuit overload | 3.4 |
| 7n59 | | 59 | 02 | PSM cooling fan stopped | 3.4 |

# 2.2 FOR SERIES 15*i*

Servo alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| SV0027 | | | | Invalid digital servo parameter setting | 3.3.6 |
| SV0360 | | | | Pulse coder checksum error (built-in) | 3.3.7 (1) |
| SV0361 | | | | Pulse coder phase error (built-in) | 3.3.7 (1) |
| SV0363 | | | | Clock error (built-in) | 3.3.7 (1) |
| SV0364 | | | | Soft phase alarm (built-in) | 3.3.7 (1) |
| SV0365 | | | | LED error (built-in) | 3.3.7 (1) |
| SV0366 | | | | Pulse error (built-in) | 3.3.7 (1) |
| SV0367 | | | | Count error (built-in) | 3.3.7 (1) |
| SV0368 | | | | Serial data error (built-in) | 3.3.7 (3) |
| SV0369 | | | | Data transfer error (built-in) | 3.3.7 (3) |
| SV0380 | | | | LED error (separate) | 3.3.7 (2) |
| SV0381 | | | | Pulse coder phase error (separate) | 3.3.7 (2) |
| SV0382 | | | | Count error (separate) | 3.3.7 (2) |
| SV0383 | | | | Pulse error (separate) | 3.3.7 (2) |
| SV0384 | | | | Soft phase alarm (separate) | 3.3.7 (2) |
| SV0385 | | | | Serial data error (separate) | 3.3.7 (3) |
| SV0386 | | | | Data transfer error (separate) | 3.3.7 (3) |
| SV0421 | | | | Excessive semi-full error | 3.3.8 |
| SV0430 | | | | Servomotor overheat | 3.3.5 |
| SV0431 | | | 03 | Converter: main circuit overload | 3.1 |
| SV0432 | | | 06 | Converter: control undervoltage/open phase | 3.1 |
| SV0433 | | | 04 | Converter: DC link undervoltage | 3.1 |
| SV0434 | 2 | | | Inverter: control power supply undervoltage | 3.2 |
| SV0435 | 5 | | | Inverter: DC link undervoltage | 3.2 |
| SV0436 | | | | Soft thermal (OVC) | 3.3.3 |
| SV0437 | | | 01 | Converter: input circuit overcurrent | 3.1 |
| SV0438 | 8<br>9<br>A<br>b<br>C<br>d<br>E | | | Inverter: current alarm (L axis)<br>(M axis)<br>(N axis)<br>(L and M axes)<br>(M and N axes)<br>(L and N axes)<br>(L, M, and N axes) | 3.2 |
| SV0439 | | | 07 | Converter: DC link overvoltage | 3.1 |
| SV0440 | | | 08 | Converter excessive deceleration power | 3.1 |
| SV0441 | | | | Current offset error | 3.3.8 |

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| SV0442 | | | 05 | Converter: DC link charging/inverter DB | 3.1 |
| SV0443 | | | 02 | Converter: cooling fan stopped | 3.1 |
| SV0444 | 1 | | | Inverter: internal cooling fan stopped | 3.2 |
| SV0445 | | | | Soft disconnection alarm | 3.3.4 |
| SV0446 | | | | Hard disconnection alarm | Not issued |
| SV0447 | | | | Hard disconnection alarm (separate) | 3.3.4 |
| SV0448 | | | | Feedback mismatch alarm | 3.3.8 |
| SV0449 | 8.<br>9.<br>A.<br>b.<br>C.<br>d.<br>E. | | | Inverter: IPM alarm (L axis)<br>(M axis)<br>(N axis)<br>(L and M axes)<br>(M and N axes)<br>(L and N axes)<br>(L, M, and N axes) | 3.2 |

Spindle alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| SP097x | | A | | Program ROM error | 3.4 |
| SP097x | | A0 | | Program ROM error | 3.4 |
| SP097x | | A1 | | Program RAM error | 3.4 |
| SP097x | | A2 | | Program RAM error | 3.4 |
| SP097x | | A3 | | SPM control circuit clock error | 3.4 |
| SP097x | | A4 | | SRAM parity error | 3.4 |
| SP0001 | | 01 | | Motor overheat | 3.4 |
| SP0002 | | 02 | | Excessive speed deviation | 3.4 |
| SP0003 | | 03 | | DC link fuse blown | 3.4 |
| SP0004 | | 04 | 06 | Power supply failure | 3.4 |
| SP0007 | | 07 | | Excessive speed | 3.4 |
| SP0009 | | 09 | | Main circuit overload | 3.4 |
| SP0011 | | 11 | 07 | DC link overvoltage | 3.4 |
| SP0012 | | 12 | | DC link overcurrent/IPM alarm | 3.4 |
| SP098x | | 13 | | CPU internal data memory error | 3.4 |
| SP0015 | | 15 | | Output switching/spindle switching alarm | 3.4 |
| SP098x | | 16 | | RAM error | 3.4 |
| SP098x | | 19 | | Excessive offset of the phase U current detection circuit | 3.4 |
| SP098x | | 20 | | Excessive offset of the phase V current detection circuit | 3.4 |
| SP022x | | 24 | | Serial transfer data error | 3.4 |
| SP022x | | 25 | | Serial data transfer stopped | 3.4 |

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| SP0026 | | 26 | | Cs contour control speed detection signal error | 3.4 |
| SP0027 | | 27 | | Position coder signal error | 3.4 |
| SP0028 | | 28 | | Cs contour control position detection signal error | 3.4 |
| SP0029 | | 29 | | Short-period overload | 3.4 |
| SP0030 | | 30 | 01 | PSM main circuit overcurrent | ‘ |
| SP0031 | | 31 | | Speed detection signal disconnected and motor lock alarm | 3.4 |
| SP098x | | 32 | | Serial communication LSI internal RAM error | 3.4 |
| SP0033 | | 33 | 05 | DC link precharge failure | 3.4 |
| SP0034 | | 34 | | Parameter data out of the specifiable range | 3.4 |
| SP0035 | | 35 | | Too large of a gear ratio is specified. | 3.4 |
| SP0036 | | 36 | | Error counter overflow | 3.4 |
| SP0037 | | 37 | | Speed detector parameter error | 3.4 |
| SP0039 | | 39 | | Cs contour control one-rotation signal detection error | 3.4 |
| SP0040 | | 40 | | Cs contour control one-rotation signal not detected | 3.4 |
| SP0041 | | 41 | | Position coder one-rotation signal detection error | 3.4 |
| SP0042 | | 42 | | Position coder one-rotation signal not detected | 3.4 |
| SP0043 | | 43 | | Differential speed mode position coder signal error | 3.4 |
| SP0044 | | 44 | | A/D conversion error | 3.4 |
| SP0046 | | 46 | | Position coder one-rotation signal detection error during thread cutting | 3.4 |
| SP0047 | | 47 | | Position coder signal error | 3.4 |
| SP0049 | | 49 | | Excessive differential speed conversion result | 3.4 |
| SP0050 | | 50 | | Excessive speed command calculation value during spindle synchronization | 3.4 |
| SP0051 | | 51 | 04 | DC link undervoltage | 3.4 |
| SP0052 | | 52 | | ITP signal error I | 3.4 |
| SP0053 | | 53 | | ITP signal error II | 3.4 |
| SP0054 | | 54 | | Current overload alarm | 3.4 |
| SP0055 | | 55 | | Power line state error at spindle switching/output switching | 3.4 |
| SP0056 | | 56 | | Control circuit cooling fan stopped | 3.4 |
| SP0056 | | 57 | 08 | Excessive regenerated power | 3.4 |
| SP0058 | | 58 | 03 | PSM main circuit overload | 3.4 |
| SP0059 | | 59 | 02 | PSM cooling fan stopped | 3.4 |

# 2.3 FOR SERIES 16, 18, 20, 21

Servo alarm

| Alarm No. | Diagnosis screen | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|---|
| 350 | Alarm 3 | | | | α pulse coder error alarm | 3.3.7 |
| 351 | Alarm 4 | | | | Pulse coder communication error alarm | 3.3.7 |
| 400 | OVL | 1 | | | Fan stopped alarm (SVM) | 3.2 |
| | | | | 2 | Fan stopped alarm (PSM) | 3.1 |
| | | | | 3 | Overheat alarm (PSM) | |
| | | | | | Overheat alarm (motor) | 3.3.5 |
| 414 | HVA | | | 7 | DC link overvoltage alarm | 3.1 |
| | HCA | 8 | | | Abnormal current alarm (L axis) | 3.2 |
| | | 9 | | | (M axis) | |
| | | A | | | (N axis) | |
| | | b | | | (L, M axis) | |
| | | C | | | (M, N axis) | |
| | | d | | | (L, N axis) | |
| | | E | | | (L, M, N axis) | |
| | | | | 1 | (PSM overcurrent + IPM alarm) | 3.1 |
| | | 8. | | | IPM alarm (L axis) | 3.2 |
| | | 9. | | | (M axis) | |
| | | A. | | | (N axis) | |
| | | b. | | | (L, M axis) | |
| | | C. | | | (M, N axis) | |
| | | d. | | | (L, N axis) | |
| | | E. | | | (L, M, N axis) | |
| | OVC | | | | Soft thermal (OVC) | 3.3.3 |
| | LVA | 2 | | | Control power supply undervoltage alarm | 3.2 |
| | | 5 | | | DC link undervoltage alarm (SVM) | |
| | | | | 4 | DC link undervoltage alarm (PSM) | 3.1 |
| | | | | 6 | Power supply state alarm | |
| | FAN | 1 | | | Fan stopped alarm (Series 20 only) | 3.2 |
| | | | | 2 | Fan stopped alarm (Series 20 only) | 3.1 |
| | MCC | | | 5 | MCC fused, precharge alarm | 3.1 |
| | OFS | | | | Current conversion error alarm | 3.3.8 |
| 416 | FBA | | | | Feedback disconnected alarm | 3.3.4 |
| 417 | PRM | | | | Invalid servo parameter setting alarm | 3.3.6 |

For the correspondence between the diagnosis screen numbers and names, see the table below.

| | DGN No. | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|---|
| Alarm 1 | 200 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |
| Alarm 2 | 201 | ALD | | | EXP | | | | |
| Alarm 3 | 202 | | CSA | BLA | PHA | RCA | BZA | CKA | SPH |
| Alarm 4 | 203 | DTE | CRC | STB | PRM | | | | |
| Alarm 5 | 204 | | OFS | MCC | LDM | PMS | FAN | DAL | ABF |

Spindle alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 749 | | A | | Program ROM error | 3.4 |
| 749 | | A0 | | Program ROM error | 3.4 |
| 749 | | A1 | | Program RAM error | 3.4 |
| 749 | | A2 | | Program RAM error | 3.4 |
| 749 | | A3 | | SPM control circuit clock error | 3.4 |
| 749 | | A4 | | SRAM parity error | 3.4 |
| 751 (AL-01) | | 01 | | Motor overheat | 3.4 |
| 751 (AL-02) | | 02 | | Excessive velocity error | 3.4 |
| 751 (AL-03) | | 03 | | DC link fuse blown | 3.4 |
| 751 (AL-04) | | 04 | 06 | Input power supply open phase and power supply failure | 3.4 |
| 751 (AL-07) | | 07 | | Overspeed | 3.4 |
| 751 (AL-09) | | 09 | | Main circuit overload | 3.4 |
| 751 (AL-11) | | 11 | 07 | DC link overvoltage | 3.4 |
| 751 (AL-12) | | 12 | | DC link overcurrent/IPM alarm | 3.4 |
| 750 | | 13 | | CPU internal data memory failure | 3.4 |
| 751 (AL-15) | | 15 | | Speed range switching and spindle switching alarm | 3.4 |
| 750 | | 16 | | RAM error | 3.4 |
| 750 | | 19 | | Phase U current detector circuit excessive offset | 3.4 |
| 750 | | 20 | | Phase V current detector circuit excessive offset | 3.4 |
| 749 | | 24 | | Serial transfer data error | 3.4 |
| 749 | | 25 | | Serial data transfer stopped | 3.4 |
| 751 (AL-26) | | 26 | | Cs contouring control velocity detection signal disconnected | 3.4 |
| 751 (AL-27) | | 27 | | Position coder signal disconnected | 3.4 |
| 751 (AL-28) | | 28 | | Cs contouring control position detection signal disconnected | 3.4 |
| 751 (AL-29) | | 29 | | Short-period overload | 3.4 |
| 751 (AL-30) | | 30 | 01 | PSM main circuit overcurrent | 3.4 |
| 751 (AL-31) | | 31 | | Speed detection signal disconnected and motor lock alarm | 3.4 |
| 750 | | 32 | | Serial communication LSI internal RAM error | 3.4 |

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 751 (AL-33) | | 33 | 05 | DC link precharge failure | 3.4 |
| 751 (AL-34) | | 34 | | Parameter data out of specification | 3.4 |
| 751 (AL-35) | | 35 | | Too large gear ratio specified | 3.4 |
| 751 (AL-36) | | 36 | | Error counter overflow | 3.4 |
| 751 (AL-37) | | 37 | | Speed detector parameter error | 3.4 |
| 751 (AL-39) | | 39 | | Cs contouring control one-rotation signal detection error | 3.4 |
| 751 (AL-40) | | 40 | | Cs contouring control one-rotation signal not detected | 3.4 |
| 751 (AL-41) | | 41 | | Position coder one-rotation signal detection error | 3.4 |
| 751 (AL-42) | | 42 | | Position coder one-rotation signal not detected | 3.4 |
| 751 (AL-43) | | 43 | | Differential speed mode position coder signal disconnected | 3.4 |
| 751 (AL-44) | | 44 | | A/D conversion error | 3.4 |
| 751 (AL-46) | | 46 | | Position coder one-rotation signal detection error during thread cutting | 3.4 |
| 751 (AL-47) | | 47 | | Position coder signal error | 3.4 |
| 751 (AL-49) | | 49 | | Excessive differential speed conversion result | 3.4 |
| 751 (AL-50) | | 50 | | Excessive speed command computed value during the synchronization control of the spindle | 3.4 |
| 751 (AL-51) | | 51 | 04 | DC link undervoltage | 3.4 |
| 751 (AL-52) | | 52 | | ITP signal error I | 3.4 |
| 751 (AL-53) | | 53 | | ITP signal error II | 3.4 |
| 751 (AL-54) | | 54 | | Overload current alarm | 3.4 |
| 751 (AL-55) | | 55 | | Power line state error at spindle or speed range switching | 3.4 |
| 751 (AL-56) | | 56 | | Control circuit cooling fan stopped | 3.4 |
| 751 (AL-57) | | 57 | 08 | Excessive regenerated power | 3.4 |
| 751 (AL-58) | | 58 | 03 | PSM main circuit overload | 3.4 |
| 751 (AL-59) | | 59 | 02 | PSM cooling fan stopped | 3.4 |

# 2.4 FOR SERIES 15

Servo alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| SV001 | | | | Soft thermal (OVC) | 3.3.3 |
| SV003 | 8 | | | Abnormal current alarm (L axis) | 3.2 |
| | 9 | | | (M axis) | |
| | A | | | (N axis) | |
| | b | | | (L, M axis) | |
| | C | | | (M, N axis) | |
| | d | | | (L, N axis) | |
| | E | | | (L, M, N axis) | |
| | | | 1 | (PSM overcurrent + IPM alarm) | 3.1 |
| | 8. | | | IPM alarm (L axis) | 3.2 |
| | 9. | | | (M axis) | |
| | A. | | | (N axis) | |
| | b. | | | (L, M axis) | |
| | C. | | | (M, N axis) | |
| | d. | | | (L, N axis) | |
| | E. | | | (L, M, N axis) | |
| SV004 | | | 7 | DC link overvoltage | 3.1 |
| SV005 | | | 5 | Precharge alarm (Series 15-A) (simultaneously with SV006) | 3.1 |
| SV006 | 2 | | | Control power supply undervoltage alarm | 3.1 |
| | 5 | | | DC link undervoltage alarm (SVM) | |
| | | | 4 | DC link undervoltage alarm (PSM) | |
| | | | 6 | Power supply state alarm | |
| | | | 5 | Precharge alarm (Series 15-A) (simultaneously with SV005) | 3.1 |
| SV015 | | | | Feedback disconnected alarm | 3.3.4 |
| SV023 | 1 | | | Fan stop alarm (SVM) | 3.2 |
| | | | 2 | Fan stop alarm (PSM) | 3.1 |
| | | | 3 | Overheat alarm (PSM) | |
| | | | | Overheat alarm (motor) | 3.3.5 |
| SV027 | | | | Invalid servo parameter setting alarm | 3.3.6 |
| SV110 | | | | α pulse coder error alarm | 3.3.7 |
| SV114 | | | | Rotation speed data error alarm | 3.3.7 |
| SV115 | | | | Pulse coder communication error alarm | 3.3.7 |
| SV116 | | | | Precharge alarm (Series 15-B) | |
| SV117 | | | | Current conversion error alarm | 3.3.8 |
| SV124 | | | | Excessive semi-full error alarm | 3.3.8 |
| SV129 | | | | Feedback mismatch alarm | 3.3.8 |

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| SV323 | | | | Soft disconnection alarm | 3.3.7 |
| SV324 | | | | Hard disconnection alarm | 3.3.7 |
| SV325 | | | | Pulse coder error alarm (built-in) | 3.3.7(1) |
| SV326 | | | | Pulse coder error alarm (built-in) | 3.3.7(1) |
| SV327 | | | | Speed error (built-in) | 3.3.7(1) |
| SV330 | | | | Serial communication error (built-in) | 3.3.7(1) |
| SV331 | | | | Pulse coder error alarm (built-in) | 3.3.7(2) |
| SV333 | | | | Pulse coder error alarm (separate linear) | 3.3.7(2) |
| SV334 | | | | Pulse coder error alarm (separate rotary) | 3.3.7(2) |
| SV335 | | | | Speed error (separate rotary) | 3.3.7(2) |
| SV338 | | | | Separate detector alarm | 3.3.7(2) |
| SV339 | | | | Serial communication error (separate) | 3.3.7(3) |
| SV340 | | | | Hard disconnection alarm (separate) | 3.3.4 |
| SV351 | | | | Soft disconnection alarm ($\alpha$ pulse coder) | 3.3.4 |

Spindle alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| OT300 | | A | | Program ROM error | 3.4 |
| OT300 | | A0 | | Program ROM error | 3.4 |
| OT300 | | A1 | | Program RAM error | 3.4 |
| OT300 | | A2 | | Program RAM error | 3.4 |
| OT300 | | A3 | | SPM control circuit clock error | 3.4 |
| OT300 | | A4 | | SRAM parity error | 3.4 |
| OT301 | | 01 | | Motor overheat | 3.4 |
| OT302 | | 02 | | Excessive velocity error | 3.4 |
| OT399 | | 03 | | DC link fuse blown | 3.4 |
| OT399 | | 04 | 06 | Power supply failure | 3.4 |
| OT307 | | 07 | | Overspeed | 3.4 |
| OT309 | | 09 | | Main circuit overload | 3.4 |
| OT311 | | 11 | 07 | DC link overvoltage | 3.4 |
| OT312 | | 12 | | DC link overcurrent/IPM alarm | 3.4 |
| OT300 | | 13 | | CPU internal data memory failure | 3.4 |
| OT399 | | 15 | | Speed range switching and spindle switching alarm | 3.4 |
| OT300 | | 16 | | RAM error | 3.4 |
| OT300 | | 19 | | Phase U current detector circuit excessive offset | 3.4 |
| OT300 | | 20 | | Phase V current detector circuit excessive offset | 3.4 |
| OT300 | | 24 | | Serial transfer data error | 3.4 |
| OT300 | | 25 | | Serial data transfer stopped | 3.4 |
| OT326 | | 26 | | Cs contouring control velocity detection signal disconnected | 3.4 |
| OT327 | | 27 | | Position coder signal disconnected | 3.4 |
| OT328 | | 28 | | Cs contouring control position detection signal disconnected | 3.4 |
| OT329 | | 29 | | Short-period overload | 3.4 |
| OT330 | | 30 | 01 | PSM main circuit overcurrent | 3.4 |
| OT331 | | 31 | | Speed detection signal disconnected and motor lock alarm | 3.4 |
| OT300 | | 32 | | Serial communication LSI internal RAM error | 3.4 |
| OT333 | | 33 | 05 | DC link precharge failure | 3.4 |
| OT334 | | 34 | | Parameter data out of specification | 3.4 |
| OT335 | | 35 | | Too large gear ratio specified | 3.4 |
| OT336 | | 36 | | Error counter overflow | 3.4 |
| OT399 | | 37 | | Speed detector parameter error | 3.4 |
| OT399 | | 39 | | Cs contouring control one-rotation signal detection error | 3.4 |

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| OT399 | | 40 | | Cs contour controling one-rotation signal not detected | 3.4 |
| OT399 | | 41 | | Position coder one-rotation signal detection error | 3.4 |
| OT399 | | 42 | | Position coder one-rotation signal not detected | 3.4 |
| OT399 | | 43 | | Differential speed mode position coder signal disconnected | 3.4 |
| OT399 | | 44 | | A/D conversion error | 3.4 |
| OT399 | | 46 | | Position coder one-rotation signal detection error during thread cutting | 3.4 |
| OT399 | | 47 | | Position coder signal error | 3.4 |
| OT399 | | 49 | | Excessive differential speed conversion result | 3.4 |
| OT399 | | 50 | | Excessive speed command computed value during the synchronization control of the spindle | 3.4 |
| OT399 | | 51 | 04 | DC link undervoltage | 3.4 |
| OT399 | | 52 | | ITP signal error I | 3.4 |
| OT399 | | 53 | | ITP signal error II | 3.4 |
| OT399 | | 54 | | Overload current alarm | 3.4 |
| OT399 | | 55 | | Power line state error at spindle or speed range switching | 3.4 |
| OT399 | | 56 | | Control circuit cooling fan stopped | 3.4 |
| OT399 | | 57 | 08 | Excessive regenerated power | 3.4 |
| OT399 | | 58 | 03 | PSM main circuit overload | 3.4 |
| OT399 | | 59 | 02 | PSM cooling fan stopped | 3.4 |

# 2.5 FOR SERIES 0-C

Servo alarm

| Alarm No. | Diagnosis screen | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|---|
| 309 | Alarm 3 | | | | α pulse coder error alarm | 3.3.7 |
| | Alarm 4 | | | | Pulse coder communication error alarm | |
| 400 | OVL | 1 | | | Fan stopped alarm (SVM) | 3.2 |
| | | | | 2 | Fan stopped alarm (PSM) | 3.1 |
| | | | | 3 | Overheat alarm (PSM) | |
| | | | | | Overheat alarm (motor) | 3.3.5 |
| 414 | HVA | | | 7 | DC link overvoltage alarm | 3.2 |
| | DCA & LVA | | | 5 | Precharge alarm | 3.2 |
| | HCA | 8 | | | Abnormal current alarm (L axis) | 3.2 |
| | | 9 | | | (M axis) | |
| | | A | | | (N axis) | |
| | | b | | | (L, M axis) | |
| | | C | | | (M, N axis) | |
| | | d | | | (L, N axis) | |
| | | E | | | (L, M, N axis) | |
| | | | | 1 | (PSM overcurrent + IPM alarm) | |
| | | 8. | | | IPM alarm (L axis) | 3.2 |
| | | 9. | | | (M axis) | |
| | | A. | | | (N axis) | |
| | | b. | | | (L, M axis) | |
| | | C. | | | (M, N axis) | |
| | | d. | | | (L, N axis) | |
| | | E. | | | (L, M, N axis) | |
| | OVC | | | | Soft thermal (OVC) | 3.3.3 |
| | LVA | 2 | | | Control power supply undervoltage alarm | 3.1 |
| | | 5 | | | DC link undervoltage alarm (SVM) | |
| | | | | 4 | DC link undervoltage alarm (PSM) | 3.2 |
| | | | | 6 | Power supply state alarm | |
| 416 | FBA | | | | Feedback disconnected alarm | 3.3.4 |
| 417 | PRM | | | | Invalid servo parameter setting alarm | 3.3.6 |

For the correspondence between the diagnosis screen numbers and names, see the table below.

| | DGN No. | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|---|
| Alarm 1 | 720 to 723 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |
| Alarm 2 | 730 to 733 | ALD | | | EXP | | | | |
| Alarm 3 | 760 to 763 | | CSA | BLA | PHA | RCA | BZA | CKA | SPH |
| Alarm 4 | 770 to 773 | DTE | CRC | STB | PRM | | | | |

Spindle alarm

| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 945 | | A | | Program ROM error | 3.4 |
| 945 | | A0 | | Program ROM error | 3.4 |
| 945 | | A1 | | Program RAM error | 3.4 |
| 945 | | A2 | | Program RAM error | 3.4 |
| 945 | | A3 | | SPM control circuit clock error | 3.4 |
| 945 | | A4 | | SRAM parity error | 3.4 |
| 409 | | 01 | | Motor overheat | 3.4 |
| 409 | | 02 | | Excessive velocity error | 3.4 |
| 409 | | 03 | | DC link fuse blown | 3.4 |
| 409 | | 04 | 06 | Input power supply open phase and power supply failure | 3.4 |
| 409 | | 07 | | Overspeed | 3.4 |
| 409 | | 09 | | Main circuit overload | 3.4 |
| 409 | | 11 | 07 | DC link overvoltage | 3.4 |
| 409 | | 12 | | DC link overcurrent/IPM alarm | 3.4 |
| 408 | | 13 | | CPU internal data memory failure | 3.4 |
| 409 | | 15 | | Speed range switching and spindle switching alarm | 3.4 |
| 408 | | 16 | | RAM error | 3.4 |
| 408 | | 19 | | Phase U current detector circuit excessive offset | 3.4 |
| 408 | | 20 | | Phase V current detector circuit excessive offset | 3.4 |
| 945 | | 24 | | Serial transfer data error | 3.4 |
| 945 | | 25 | | Serial data transfer stopped | 3.4 |
| 409 | | 26 | | Cs contouring control velocity detection signal disconnected | 3.4 |
| 409 | | 27 | | Position coder signal disconnected | 3.4 |
| 409 | | 28 | | Cs contouring control position detection signal disconnected | 3.4 |
| 409 | | 29 | | Short-period overload | 3.4 |
| 409 | | 30 | 01 | PSM main circuit overcurrent | 3.4 |
| 409 | | 31 | | Speed detection signal disconnected and motor lock alarm | 3.4 |
| 408 | | 32 | | Serial communication LSI internal RAM error | 3.4 |
| 409 | | 33 | 05 | DC link precharge failure | 3.4 |
| 409 | | 34 | | Parameter data out of specification | 3.4 |
| 409 | | 35 | | Too large gear ratio specified | 3.4 |
| 409 | | 36 | | Error counter overflow | 3.4 |
| 409 | | 37 | | Speed detector parameter error | 3.4 |
| 409 | | 39 | | Cs contouring control one-rotation signal detection error | 3.4 |

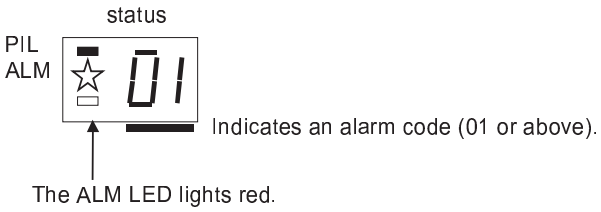
| Alarm No. | SVM | SPM | PSM | Description | Remarks |
|---|---|---|---|---|---|
| 409 | | 40 | | Cs contouring control one-rotation signal not detected | 3.4 |
| 409 | | 41 | | Position coder one-rotation signal detection error | 3.4 |
| 409 | | 42 | | Position coder one-rotation signal not detected | 3.4 |
| 409 | | 43 | | Differential speed mode position coder signal disconnected | 3.4 |
| 409 | | 44 | | A/D conversion error | 3.4 |
| 409 | | 46 | | Position coder one-rotation signal detection error during thread cutting | 3.4 |
| 409 | | 47 | | Position coder signal error | 3.4 |
| 409 | | 49 | | Excessive differential speed conversion result | 3.4 |
| 409 | | 50 | | Excessive speed command computed value during the synchronization control of the spindle | 3.4 |
| 409 | | 51 | 04 | DC link undervoltage | 3.4 |
| 409 | | 52 | | ITP signal error I | 3.4 |
| 409 | | 53 | | ITP signal error II | 3.4 |
| 409 | | 54 | | Overload current alarm | 3.4 |
| 409 | | 55 | | Power line state error at spindle or speed range switching | 3.4 |
| 409 | | 56 | | Control circuit cooling fan stopped | 3.4 |
| 409 | | 57 | 08 | Excessive regenerated power | 3.4 |
| 409 | | 58 | 03 | PSM main circuit overload | 3.4 |
| 409 | | 59 | 02 | PSM cooling fan stopped | 3.4 |

# 3 TROUBLESHOOTING AND ACTION

# 3.1 POWER SUPPLY MODULE

## 3.1.1 Power Supply Module (PSM, PSM-HV)

If an alarm occurs, in the STATUS display, the ALM LED lights red, and the two-digit 7-segment display indicates an alarm code.




### 3.1.1.1 Alarm code 01

For the PSM-5.5 to PSM-11

(1) Meaning
The main circuit power module (IPM) has detected an overload, overcurrent, or control supply voltage decrease.

(2) Cause and troubleshooting
(a) Cooling fan failure
Check whether the cooling fan rotates normally.
→ Replace the cooling fan.
(b) Dust buildup
→ Clean the cooling system with a vacuum cleaner or compressed air.
(c) Overload
→ Examine the operating conditions.
(d) Input supply voltage imbalance
→ Check the input power supply specification.
(e) The specification of the AC reactor does not match the PSM in use.
→ Check the PSM and the specification of the AC reactor.
(f) IPM failure, or control supply voltage decrease of the power module (IPM)
→ Replace the unit or IPM.

For PSM-15 to PSM-55, PSM-18HV to PSM-75HV

(1) Meaning
Overcurrent flowed into the input of the main circuit.

(2) Cause and troubleshooting
(a) Input supply voltage imbalance
→ Check the input power supply specification.
(b) The specification of the AC reactor does not match the PSM in use.
→ Check the PSM and the specification of the AC reactor.
(c) IGBT defective
→ Replace IGBT.

### 3.1.1.2 Alarm code 02

(1) Meaning
(a) A cooling fan for the control circuit has stopped.
(b) The control supply voltage has dropped.

(2) Cause and troubleshooting
(a) Cooling fan broken
Check whether the cooling fan rotates normally.
→ Replace it.
(b) Input voltage decrease
→ Check the power supply.

### 3.1.1.3 Alarm code 03

(1) Meaning
The temperature of the main circuit heat sink has risen abnormally.

(2) Cause and troubleshooting
(a) Cooling fan for the main circuit broken
Check whether the cooling fan for the main circuit rotates normally.
→ Replace it.
(b) Dust accumulation
→ Clean the cooling system with a vacuum cleaner or the factory air blower.
(c) Overload
→ Examine the operating conditions.

### 3.1.1.4 Alarm code 04

(1) Meaning
In the main circuit, the DC voltage (DC link) has dropped.

(2) Cause and troubleshooting
(a) A small power dip has occurred.
→ Check the power supply.
(b) Low input power supply voltage
→ Check the power supply specification.
(c) The main circuit power supply may have been switched off with an emergency stop state released.
→ Check the sequence.

### 3.1.1.5 Alarm code 05

(1) Meaning
(a) The input power supply is abnormal (open phase).
(b) The main circuit capacitor was not recharged within the specified time.

(2) Cause and troubleshooting
(a) The input power supply has an open phase.
→ Check the connection.
(b) Too many SVM and/or SPM units are connected.
→ Check the specification of the PSM.
(c) The DC link is short-circuited.
→ Check the connection.
(d) The recharge current limiting resistor is defective.
→ Replace the unit.

### 3.1.1.6 Alarm code 06

(1) Meaning
The input power supply is abnormal (open phase).

(2) Cause and troubleshooting
(a) The input power supply has an open phase.
→ Check the connection.

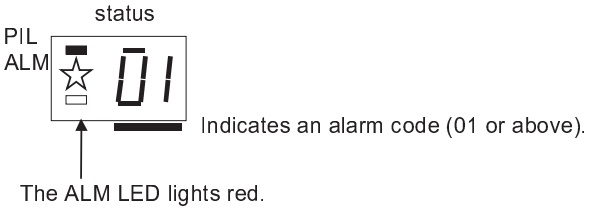
### 3.1.1.7 Alarm code 07

(1) Meaning
In the main circuit, the DC voltage at the DC link is abnormally high.

(2) Cause and troubleshooting
(a) Excessive regenerated power
Regeneration is impossible. The PSM does not have a sufficient capacity.
→ Check the specification of the PSM.
(b) The output impedance of the AC power source is too high.
→ Check the power source output impedance.
(Normal if the voltage variation at maximum output time is within 7%)
(c) Regeneration circuit failure
→ Replace the unit.
(d) The main circuit power supply may have been switched off with an emergency stop state released.
→ Check the sequence.

## 3.1.2 Power Supply Module (PSMV)

If an alarm occurs, in the STATUS display, the ALM LED lights red, and the two-digit 7-segment display indicates an alarm code.



### 3.1.2.1 Alarm code 01

(1) Meaning
Overcurrent flowed into the DC link of the main circuit.

(2) Cause and troubleshooting
(a) Thyristor or IGBT failure
→ Replace the thyristor or IGBT.
(b) DC link short-circuit
→ Check the connection.
(c) The specification of the AC reactor does not match the PSMV in use.
→ Check the PSMV and the specification of the AC reactor.

### 3.1.2.2 Alarm code 02

(1) Meaning
A cooling fan for the control circuit has stopped.

(2) Cause and troubleshooting
(a) Cooling fan for the control circuit broken
Check whether the cooling fan for the control circuit rotates normally.
→ Replace it.

### 3.1.2.3 Alarm code 03

(1) Meaning
The temperature of the main circuit heat sink has risen abnormally.

(2) Cause and troubleshooting
(a) Cooling fan for the main circuit broken
Check whether the cooling fan for the main circuit rotates normally.
→ Replace it.
(b) Dust accumulation
→ Clean the cooling system with a vacuum cleaner or the factory air blower.
(c) Overload
→ Examine the operating conditions.

### 3.1.2.4 Alarm code 04

(1) Meaning
In the main circuit, the DC voltage (DC link) has dropped.

(2) Cause and troubleshooting
(a) Low input power supply voltage
→ Check the power supply.

### 3.1.2.5 Alarm code 05

(1) Meaning
The main circuit capacitor was not recharged within the specified time.

(2) Cause and troubleshooting
(a) Too many SVM and/or SPM units are connected.
→ Check the specification of the PSMV.

### 3.1.2.6 Alarm code 06

(1) Meaning
(a) The main circuit supply voltage is not fed.
(b) Two or more input fuses of the AC reactor unit have blown.
Note) This alarm is also output when a momentary power failure occurs.

(2) Cause and troubleshooting
(a) The input supply voltage is not fed to the main circuit.
→ Check the connection.
(b) Two or more input fuses of the AC reactor unit have blown.
→ Replace the fuses.

### 3.1.2.7 Alarm code 16

(1) Meaning
(a) The main circuit power supply has an open phase.
(b) One input fuse of the AC reactor unit has blown.

(2) Cause and troubleshooting
(a) The main circuit power supply has an open phase.
→ Check the connection.
(b) One input fuse of the AC reactor unit has blown.
→ Replace the fuse.

### 3.1.2.8 Alarm code 26

(1) Meaning
The frequency of the main circuit input power supply is abnormal.

(2) Cause and troubleshooting
(a) Input power supply frequency error
→ Check the input power supply specification (50/60 Hz ±1 Hz).

### 3.1.2.9 Alarm code 36

(1) Meaning
The input power supply of the main circuit has an imbalance.

(2) Cause and troubleshooting
(a) The input supply voltage has an imbalance of 5% or more.
→ Check the input power supply specification.

### 3.1.2.10 Alarm code 46

(1) Meaning
When the magnetic contactor is turned on, the phase sequence of the power supply cannot be determined.

(2) Cause and troubleshooting
(a) Input power supply error
→ Check the input power supply specification.

### 3.1.2.11 Alarm code 07

(1) Meaning
In the main circuit, the DC voltage at the DC link is abnormally high.

(2) Cause and troubleshooting
(a) Excessive regenerated power
The PSMV does not have a sufficient capacity.
→ Check the specification of the PSMV.
(b) The output impedance of the AC power source is too high.
→ Check the power source output impedance.
(Normal if the voltage variation at maximum output time is within 7%)
(c) Regeneration circuit failure
→ Replace the unit.

## 3.1.2.12 Alarm code 17

(1) Meaning
In the main circuit, the voltage at the DC link is abnormally high. A fuse of the AC reactor unit may have blown.

(2) Cause and troubleshooting
(a) An input fuse of the AC reactor unit has blown.
→ Replace the fuse.
(b) Regeneration circuit failure
→ Replace the unit.

## 3.1.2.13 Alarm code 08

(1) Meaning
The offset of the current detection circuit of the main circuit DC link is excessive.

(2) Cause and troubleshooting
(a) The current detection circuit of the main circuit DC link malfunctions.
→ Replace the power printed wiring board.
(b) The control printed circuit board or A/D converter is faulty.
→ Replace the control printed circuit board.

## 3.1.2.14 Alarm code 18

(1) Meaning
An error occurred in internal parameter data transfer processing.

(2) Cause and troubleshooting
(a) The control printed circuit board is faulty.
→ Replace the control printed circuit board.

## 3.1.2.15 Alarm code A0

(1) Meaning
  (a) No ROM is installed.
  (b) The ROM is faulty.

(2) Cause and troubleshooting
  (a) The ROM is not installed correctly, or no ROM is installed.
  Check if the ROM is removed from the socket or if a bent lead has caused a bad contact.
  → Install the ROM correctly.
  (b) ROM specification error
  Check the software version stamped on the ROM.
  → Install a specified ROM correctly.
  (c) Control printed circuit board failure
  → Replace the control printed circuit board.

## 3.1.2.16 Alarm code A1

(1) Meaning
The RAM is faulty.

(2) Cause and troubleshooting
  (a) The control printed circuit board is faulty.
  → Replace the control printed circuit board.

## 3.1.2.17 Alarm code A2

(1) Meaning
A program is not operating normally.

(2) Cause and troubleshooting
  (a) The control printed circuit board is faulty.
  → Replace the control printed circuit board.

## 3.1.3 Power Supply Module (PSMR)

When an alarm is issued, the one-digit 7-segment LED indicates an alarm code.

Indicates an alarm code (02 or above).

### 3.1.3.1 Alarm code 2

(1) Meaning
A cooling fan for the control circuit has stopped.

(2) Cause and troubleshooting
(a) Cooling fan for the control circuit broken
Check whether the cooling fan for the control circuit rotates normally.
→ Replace it.

### 3.1.3.2 Alarm code 4

(1) Meaning
In the main circuit, the DC voltage (DC link) has dropped.

(2) Cause and troubleshooting
(a) A small power dip has occurred.
→ Check the power supply.
(b) Low input power supply voltage
→ Check the power supply specification.
(c) The main circuit power supply may have been switched off with an emergency stop state released.
→ Check the sequence.

### 3.1.3.3 Alarm code 5

(1) Meaning
(a) The input power supply is abnormal (open phase).
(b) The main circuit capacitor was not recharged within the specified time.

(2) Cause and troubleshooting
(a) The input power supply has an open phase.
→ Check the connection.
(b) Too many SVM and/or SPM units are connected.
→ Check the specification of the PSM.
(c) The DC link is short-circuited.
→ Check the connection.
(d) The recharge current limiting resistor is defective.
→ Replace the unit.

### 3.1.3.4 Alarm code 6

(1) Meaning
The control supply voltage has dropped.

(2) Cause and troubleshooting
(a) Input voltage decrease
→ Check the power supply.

### 3.1.3.5 Alarm code 7

(1) Meaning
In the main circuit, the DC voltage at the DC link is abnormally high.

(2) Cause and troubleshooting
(a) Excessive regenerated power
The PSMR does not have a sufficient capacity.
→ Check the specification of the PSMR.
(b) Regeneration circuit failure
→ Replace the unit.

## 3.1.3.6 Alarm code 8

(1) Meaning
  (a) The regenerative discharge unit is heated.
  (b) Regenerated power is excessive.

(2) Cause and troubleshooting
  (a) The regenerative resistance capacity is insufficient.
    → Check the specification of the regenerative resistance.
  (b) Excessive regenerated power
    → Decrease the frequency of acceleration/deceleration during operation.
  (c) The cooling fan of the regenerative discharge unit is faulty.
    → Check the rotation state of the cooling fan.

# 3.2 SERVO AMPLIFIER MODULE

## 3.2.1 Abnormal Current Alarms (8, 9, A, b, C, d, and E in the LED display)

(1) Make sure that the following parameters are set to the standard values. If they are not, abnormal current control is performed.

| No. 1809 | |
|---|---|
| No. 2004 | No. 8X04 |

| No. 1884 | |
|---|---|
| No. 2006 | No. 8X06 |

| No. 1954(15-A),1955(15-B) | |
|---|---|
| No. 2011 | No. 8X10 |

| No. 1852 | |
|---|---|
| No. 2040 | No. 8X40 |

| No. 1853 | |
|---|---|
| No. 2041 | No. 8X41 |

| No. 1967 | |
|---|---|
| No. 2074 | No. 8X74 |

| No. 1991 | |
|---|---|
| No. 2098 | No. 8X98 |

(2) Remove the power line wires from the amplifier terminals, and release an emergency stop state.
→ If an abnormal current alarm is issued, the servo amplifier module needs to be replaced.
→ If not, go to (3).

(3) Check for insulation between PE and each of the removed power wires U, V, and W. If insulation is perfect, go to (4). If not, disconnect the power wires from the motor connector. Then check for insulation between PE and each of the U, V, and W terminals on the motor.
→ If there is a short-circuit between PE and U, V, or W of the motor, replace the motor.
→ If insulation is perfect, replace the power wires.

(4) Connect the power wires. Attach the check board (A06B-6071-K290) to connector JX5 to measure the waveform of the actual current (IR and IS) in the servo amplifier module. Accelerate or decelerate the motor, and measure the actual current (IR and IS) of the amplifier.
→ If an abnormal current alarm occurs right after an emergency stop state is released, go to (5).

Release an emergency stop state, and start the motor.
Check whether the waveform of the actual current (IR and IS) is a normal sine wave.
→ If normal, go to (5).
→ If not, replace the amplifier.

(5) Check whether there is noise on the actual current (IR and IS) waveform.
→ If there is no noise, replace the amplifier.
→ If there is noise, use a shielding wire, and ground the shielding, or take other countermeasures as required.

(6) If still there is noise, a probable cause is a defective command cable or a hardware failure in the CNC.

## 3.2.2 IPM Alarms (8., 9., A., b., C., d., and E in the LED display; note these codes are displayed simultaneously with a period.)

(1) Wait for about 10 minutes. Then release the emergency stop state. If an IPM alarm still occurs, go to (2).
If the cause is IPM overheat, the IPM alarm will not recur. IPM overheat can occur if the ambient temperature is high or the motor is overloaded. Check the operating condition.

(2) Remove the power wires from the amplifier terminals, and release an emergency stop state.
If the IPM alarm does not recur, go to (3).

If the IPM alarm recurs, the probable cause is the operation of the IPM protective function (for overcurrent or power failure). Replace the amplifier and see.
→ If the IPM does not recur, go to (3).

(3) Check for insulation between PE and each of the removed power wires U, V, and W. If insulation is perfect, go to (4). If not, disconnect the power wires from the motor connector. Then check for insulation between PE and each of the U, V, and W terminals on the motor.
→ If there is a short-circuit between PE and U, V, or W of the motor, replace the motor.
→ If insulation is perfect, replace the power wires.

(4) Connect the power wires. Attach the check board (A06B-6071-K290) to connector JX5 to measure the waveform of the actual current (IR and IS) in the servo amplifier module. Accelerate or decelerate the motor, and measure the actual current (IR and IS) of the amplifier.
If an overcurrent alarm occurs right after an emergency stop state is released, go to (5).
Release an emergency stop state, and start the motor.
Check whether the waveform of the actual current (IR and IS) is a normal sine wave.
→ If normal, go to (5).
→ If not, replace the amplifier.

(5) Check whether there is noise on the actual current (IR and IS) waveform.
→ If there is no noise, replace the amplifier.
→ If there is noise, use a shielding wire, and ground the shielding, or take other countermeasures as required.

(6) If still there is noise, a probable cause is a defective command cable or a hardware failure in the CNC.

## 3.2.3 Control Power Supply Undervoltage Alarm (2 in the LED display)

(1) Check the three-phase input voltage to the amplifier.
→ If the voltage is below 0.85 times the rating, adjust it to the rated value.

(2) Replace the servo amplifier.

## 3.2.4 DC Link Undervoltage Alarm (5 in the LED display)

(1) Check the three-phase input voltage to the power supply module.
→ If the voltage is below 0.85 times the rating, adjust it to the rated value.

(2) Replace the servo amplifier module.

## 3.2.5 Fan Stopped Alarm (1 in the LED display)

(1) Make sure that the fan is not clogged up.

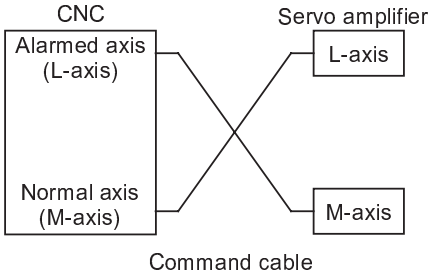
(2) Replace the servo amplifier module. Alternatively, replace the cooling fan inside the servo amplifier module according to Chapter 4.

## 3.2.6 Current Conversion Error Alarm

(1) Exchange the command cable with the cable for the axis on which no alarm has occurred.
→ If the alarm occurs on the same axis, go to (3).
→ If the alarm occurs on the new axis, go to (2).

(2) The command cable is defective. Replace it.

(3) Exchange the command cables according to the diagram here. When switching the CNC on, do so in an emergency stop state.
→ If the alarm recurs on the same axis, go to (5).
→ If the alarm occurs on the other axis, go to (4).




(4) The servo amplifier is defective.

(5) The module for current conversion in the CNC is defective.

# 3.3 SERVO SOFTWARE

If a servo alarm is issued, an alarm message is output, and details of the alarm are also displayed on the servo adjustment screen or the diagnosis screen. Using the alarm identification table given in this section, determine the alarm, and take a proper action. With a CNC of the *i* series, detailed alarm messages are displayed. So, from the information in the alarm identification table, select a proper action.

## 3.3.1 Servo Adjustment Screen

- **Series 0-C**

Press the [PARAM] key several times to display the servo setting screen.

Pressing the [↑ PAGE] [PAGE ↓] keys displays the servo screen.

If the servo setting screen does not appear, specify the following parameter, then switch the CNC off and on again:

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| 0389 | | | | | | | | SVS |

SVS (#0)=0 (to display the servo setting screen)

- **Series 15-A/B, 15*i***

Press the [SERVICE] key several times to cause the servo setting screen to appear. Then press the [↓] key, and the servo adjustment screen will appear.

- **Series 16, 18, 20, 21**

[SYSTEM] → [SYSTEM] → [ ▷] → [SV-PRM] → [SV-TUN]

If the servo setting screen does not appear, specify the following parameter, then switch the CNC off and on again.

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| 3111 | | | | | | | | SVS |

SVS (#0)=1 (to display the servo setting screen)

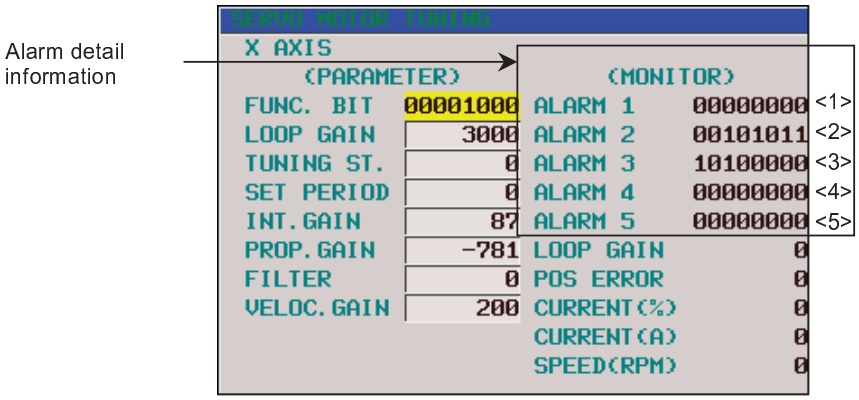



**Fig. 3.3.1(a) Servo adjustment screen**

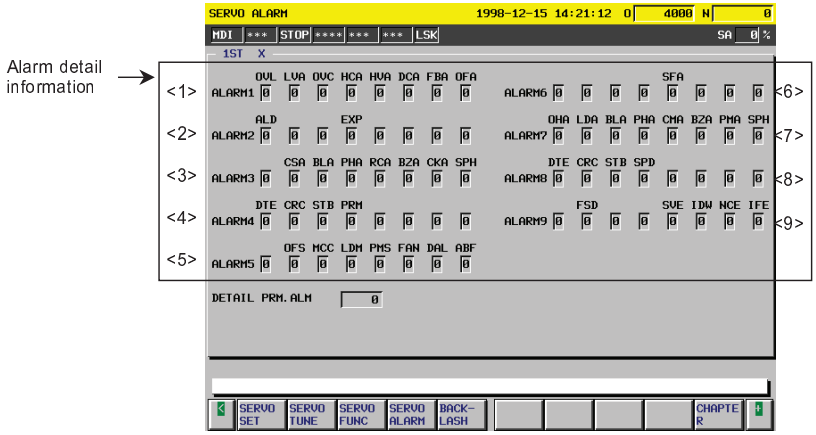



**Fig. 3.3.1(b) Series 15*i* servo alarm screen**

The table below indicates the names of the alarm bits.

**Table 3.3.1 List of alarm bit names**

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <1> Alarm 1 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |
| <2> Alarm 2 | ALD | | | EXP | | | | |
| <3> Alarm 3 | | CSA | BLA | PHA | RCA | BZA | CKA | SPH |
| <4> Alarm 4 | DTE | CRC | STB | PRM | | | | |
| <5> Alarm 5 | | OFS | MCC | LDM | PMS | FAN | DAL | ABF |
| <6> Alarm 6 | | | | | SFA | | | |
| <7> Alarm 7 | OHA | LDA | BLA | PHA | CMA | BZA | PMA | SPH |
| <8> Alarm 8 | DTE | CRC | STB | SPD | | | | |
| <9> Alarm 9 | | FSD | | | SVE | IDW | NCE | IFE |

**NOTE**
The empty fields do not represent alarm codes.

## 3.3.2 Diagnosis Screen

The alarm items of the servo adjustment screen correspond to the diagnosis screen numbers indicated in the table below.

**Table 3.3.2 Correspondence between the servo adjustment screen and diagnosis screen**

| Alarm No. | Series 0-C | Series 15-A, B, 15*i* | Series 16, 18, 20, 21 | PowerMate-E |
|---|---|---|---|---|
| <1> Alarm 1 | No 720 to 723 | No 3014 + 20(X-1) | No 200 | No 2711 |
| <2> Alarm 2 | 730 to 733 | 3015 + 20(X-1) | 201 | 2710 |
| <3> Alarm 3 | 760 to 763 | 3016 + 20(X-1) | 202 | 2713 |
| <4> Alarm 4 | 770 to 773 | 3017 + 20(X-1) | 203 | 2712 |
| <5> Alarm 5 | ———— | ———— | 204 | 2714 |
| <6> Alarm 6 | ———— | ———— | ——— | ——— |
| <7> Alarm 7 | ———— | ———— | 205 | ——— |
| <8> Alarm 8 | ———— | ———— | 206 | ——— |
| <9> Alarm 9 | ———— | ———— | ——— | ——— |

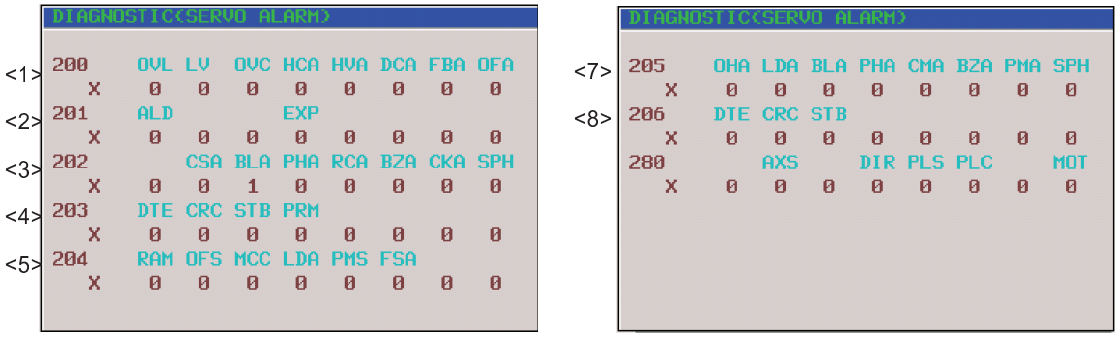



**Fig. 3.3.2 Series 16*i* diagnosis screen**

## 3.3.3 Overload Alarm (Soft Thermal, OVC)

(Alarm identification method)

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <1> Alarm 1 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |

(Action)

(1) Make sure that the motor is not vibrating.
⇒ If a motor vibrates, the current flowing in it becomes more than necessary, resulting in an alarm.

(2) Make sure that the power line to the motor is connected correctly.
⇒ If the connection is incorrect, an abnormal current flows in the motor, resulting in an alarm.

(3) Make sure that the following parameters are set correctly.
⇒ An overload alarm is issued based on the result of calculation of these parameters. Be sure to set them to the standard values. For details of the standard values, refer to the parameter manual (B-65150E).

| | | |
|---|---|---|
| 1877 | 8x62 | Overload protection coefficient (OVC1) |
| 2062 | 1062 | |

| | | |
|---|---|---|
| 1878 | 8x63 | Overload protection coefficient (OVC2) |
| 2063 | 1063 | |

| | | |
|---|---|---|
| 1893 | 8x65 | Overload protection coefficient (OVCLMT) |
| 2065 | 1065 | |

(4) Attach the check board (A06B-6071-K290) to connector JX5 to measure the waveform of the actual current (IR and IS) of the servo amplifier module. Start the motor and measure the actual current (IR and IS).
⇒ If the actual current exceeds 1.4 times the rated current, the constant for the acceleration/deceleration duration is too small, or the load on the machine is too heavy for the capacity of the motor.
⇒ If the actual current exceeds 1.4 times the rated current during normal operation, the load on the machine is too heavy for the capacity of the motor.

## 3.3.4 Feedback Disconnected Alarm

(Alarm identification method)

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <1> Alarm 1 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |
| <2> Alarm 2 | ALD | | | EXP | | | | |
| <6> Alarm 6 | | | | | SFA | | | |

| FBA | ALD | EXP | SFA | Alarm description | Action |
|---|---|---|---|---|---|
| 1 | 1 | 1 | 0 | Hard disconnection (separate phase A/B) | 1 |
| 1 | 0 | 0 | 0 | Soft disconnection (closed loop) | 2 |
| 1 | 0 | 0 | 1 | Soft disconnection (α pulse coder) | 3 |

(Action)

**Action 1:** This alarm is issued when a separate phase A/B scale is used. Check if the phase A/B detector is connected correctly.

**Action 2:** This alarm is issued when the position feedback pulse variation is small relative to the velocity feedback pulse variation. This means that this alarm is not issued when a semi-full is used. Check if the separate detector outputs position feedback pulses correctly. If position feedback pulses are output correctly, it is considered that the motor alone is rotating in the reverse direction at the start of machine operation because of a large backlash between the motor position and scale position.

| | | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|---|
| 1808 | 8X03 | | | | | | | TGAL | |
| 2003 | 1003 | | | | | | | | |

TGAL (#1) 1: Uses the parameter for the soft disconnection alarm detection level.

| | | |
|---|---|---|
| 1892 | 8X64 | Soft disconnection alarm level |
| 2064 | 1064 | |

Standard setting 4: Alarm issued for a 1/8 rotation of the motor. Increase this value.

**Action 3:** This alarm is issued when synchronization between the absolute position data sent from the built-in pulse coder and phase data is lost. Turn off the power to the CNC, then detach the pulse coder cable then attach it again. If this alarm is still issued, replace the pulse coder.

## 3.3.5 Overheat Alarm

(Alarm identification method)

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <1> Alarm 1 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |
| <2> Alarm 2 | ALD | | | EXP | | | | |

| OVL | ALD | EXP | Alarm description | Action |
|---|---|---|---|---|
| 1 | 1 | 0 | Motor overheat | 1 |
| 1 | 0 | 0 | Amplifier overheat | 1 |

(Action)

**Action 1:** If this alarm is issued after a long-time of continuous operation, it is considered that the motor and amplifier are overheated. Stop operation for a while, then make a check. If this alarm is still issued after the power is off for about 10 minutes then is turned on again, the thermostat is considered to be faulty. If this alarm is issued intermittently, increase the time constant or increase stop time in the program to suppress the rise in temperature.

## 3.3.6 Invalid Servo Parameter Setting Alarm

The invalid servo parameter setting alarm is issued when a setting out of the specifiable range is specified, or an overflow has occurred in an internal calculation. When an invalid parameter is detected on the servo side, alarm 4 #4 (PRM) = 1 results.
(Alarm identification method)

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <4> Alarm 4 | DTE | CRC | STB | PRM | | | | |

For details and action required when the invalid servo parameter setting alarm is issued on the servo side, refer to the parameter manual (B-65150E) edition 4 and up.

(Reference information)

Method of checking details of an invalid parameter detected on the servo side

(For Series 15*i*)

A number is indicated in the item "Details of invalid parameter" on the servo alarm screen (Fig. 3.3.1(b)).

(For Series 16*i*, 18*i*, 21*i*, and Power Mate *i*)

A number is indicated in No. 352 of the diagnosis screen.

## 3.3.7 Alarms Related to Pulse Coder and Separate Serial Detector

(Bits for alarm identification)

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <1> Alarm 1 | OVL | LVA | OVC | HCA | HVA | DCA | FBA | OFA |
| <2> Alarm 2 | ALD | | | EXP | | | | |
| <3> Alarm 3 | | CSA | BLA | PHA | RCA | BZA | CKA | SPH |
| <4> Alarm 4 | DTE | CRC | STB | PRM | | | | |
| <5> Alarm 5 | | OFS | MCC | LDM | PMS | FAN | DAL | ABF |
| <6> Alarm 6 | | | | | SFA | | | |
| <7> Alarm 7 | OHA | LDA | BLA | PHA | CMA | BZA | PMA | SPH |
| <8> Alarm 8 | DTE | CRC | STB | SPD | | | | |
| <9> Alarm 9 | | FSD | | | SVE | IDW | NCE | IFE |

(1) For a built-in pulse coder
An alarm is determined from the bits of alarms 1, 2, 3, and 5. The table below indicates the meaning of each bit.

| Alarm 3 | | | | | | | Alarm 5 | | 1 | Alarm 2 | | Alarm description | Action |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CSA | BLA | PHA | RCA | BZA | CKA | SPH | LDM | PMA | FBA | ALD | EXP | | |
| | | | | | | 1 | | | | | | Soft phase alarm | 2 |
| | | | | | 1 | | | | | | | Clock alarm (serial A) | |
| | | | | 1 | | | | | | | | Zero battery voltage | 1 |
| | | | 1 | | | | | | 0 | 0 | 0 | Speed error (serial A) | |
| | | | 1 | | | | | | 1 | 1 | 0 | Count error alarm (α pulse coder) | 2 |
| | | 1 | | | | | | | | | | Phase alarm (serial A) | 2 |
| | 1 | | | | | | | | | | | Battery voltage decrease (warning) | 1 |
| 1 | | | | | | | | | | | | Checksum alarm (serial A) | |
| | | | | | | | | 1 | | | | Pulse error alarm (α pulse coder) | |
| | | | | | | | 1 | | | | | LED error alarm (α pulse coder) | |

**NOTE**
An alarm for which no action number is given is considered to be caused by a pulse coder failure. Replace the pulse coder.

(2) For a separate serial detector
An alarm is determined from the bits of alarm 7. The table below indicates the meaning of each bit.

| Alarm 7 | | | | | | | | Alarm description | Action |
|---|---|---|---|---|---|---|---|---|---|
| OHA | LDA | BLA | PHA | CMA | BZA | PMA | SPH | | |
| | | | | | | | 1 | Soft phase alarm | 2 |
| | | | | | | 1 | | Pulse error alarm | |
| | | | | | 1 | | | Zero battery voltage | 1 |
| | | | | 1 | | | | Count error alarm | 2 |
| | | | 1 | | | | | Phase alarm | 2 |
| | | 1 | | | | | | Battery voltage decrease (warning) | 1 |
| | 1 | | | | | | | LED error alarm | |
| 1 | | | | | | | | Separate detector alarm | |

**NOTE**
An alarm for which no action number is given is considered to be caused by a detector failure. Replace the detector.

(Action)

**Action 1: Battery-related alarms**
Check if a battery is connected. When the power is turned on for the first time after a battery is connected, the zero battery voltage alarm is issued. In such a case, turn off the power, then turn on the power again. If the alarm is still issued, check the battery voltage. If the battery voltage decrease alarm is issued, check the voltage, and replace the battery as required.

**Action 2: Alarms that may be issued for noise**
If an alarm is issued intermittently or after emergency stop cancellation, noise is probably the cause. So, provide noise protection. If the same alarm is still issued after noise protection is provided, replace the detector.

(3) Alarms related to serial communication
An alarm is determined from the bits of alarms 4 and 8.

| Alarm 4 | | | Alarm 8 | | | Alarm description |
|---|---|---|---|---|---|---|
| DTE | CRC | STB | DTE | CRC | STB | |
| 1 | | | | | | Serial pulse coder communication alarm |
| | 1 | | | | | |
| | | 1 | | | | |
| | | | 1 | | | Separate serial pulse coder communication alarm |
| | | | | 1 | | |
| | | | | | 1 | |

**Action:** Serial communication is not performed correctly. Check if the cable is connected correctly and is not broken. If CRC or STB is issued, noise may be the cause. So, provide noise protection. If CRC or STB is always issued after the power is turned on, the pulse coder or amplifier control board (*i* series) or the pulse module (*i* series) may be faulty.

## 3.3.8 Other Alarms

(Alarm identification method)

| | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|
| <5> Alarm 5 | | OFS | MCC | LDM | PMS | FAN | DAL | ABF |

| OFS | DAL | ABF | Alarm description | Action |
|---|---|---|---|---|
| | | 1 | Feedback mismatch alarm | 1 |
| | 1 | | Excessive semi-full error alarm | 2 |
| 1 | | | Current offset error alarm | 3 |

(Action)

**Action 1:** This alarm is issued when the move direction of the position detector is opposite to the move direction of the speed detector. Check the rotation direction of the separate detector. If the rotation direction of the separate detector is opposite to the rotation direction of the motor, take the following action:

For a phase A/B detector: Reverse the connections of A and $\overline{A}$.

For a serial detector: Reverse the setting of the signal direction of the separate detector.

| | | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|---|
| 1960 | - | | | | | | | | RVRSE |
| 2018 | - | | | | | | | | |

RVRSE (#0) Reverses the signal direction of the separate detector.
0: Does not reverse the signal direction of the separate detector.
1: Reverses the signal direction of the separate detector.

If a large distortion exists between the motor and separate detector, this alarm may be issued in the case of abrupt acceleration/ deceleration. In such a case, modify the detection level.

| | | #7 | #6 | #5 | #4 | #3 | #2 | #1 | #0 |
|---|---|---|---|---|---|---|---|---|---|
| 1741 | - | | | | | | | RNLV | |
| 2201 | - | | | | | | | | |

RNLV (#1) Modifies the feedback mismatch alarm detection level.
1: Detected with 1000 $min^{-1}$ or more
0: Detected with 600 $min^{-1}$ or more

**Action 2:** This alarm is issued when the difference between the motor position and separate detector position exceeds the excessive semi-full error level. Check if the conversion efficient for dual position feedback is set correctly. If the conversion efficient is set correctly, increase the alarm level. If this alarm is still issued after the level is modified, check the connection direction of the scale.

| | | |
|---|---|---|
| 1971 | - | Dual position feedback conversion coefficient (numerator) |
| 2078 | - | |

| | | |
|---|---|---|
| 1972 | - | Dual position feedback conversion coefficient (denominator) |
| 2079 | - | |

$$\text{Conversion coefficient} = \frac{\left(\text{Number of feedback pulses per motor revolution (detection unit)}\right)}{1{,}000{,}000}$$

| | | |
|---|---|---|
| 1729 | - | Dual position feedback semi-full error level |
| 2118 | - | |

[Setting] Detection unit. When 0 is set, no detection is made.

**Action 3:** The current offset value of the current detector (equivalent to the current value in the emergency stop state) is abnormally high. If this alarm is still issued after the power is turned off then back on, the current detector is faulty. For the *i* series, replace the control board of the amplifier. For series other than the *i* series, replace the servo-related modules on the CNC.

# 3.4 SPINDLE AMPLIFIER MODULE

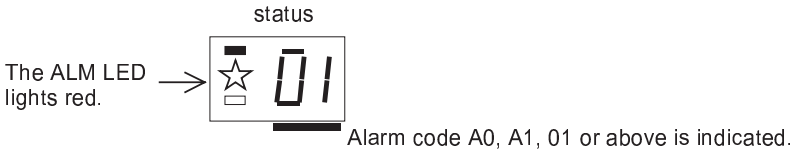
This section uses the following format for describing parameter numbers:

| FS0 | FS15*i* | FS16*i*/16, PM-D/F | Description |
|---|---|---|---|
| 6582 | 3082 | 4082 | Setting of acceleration/deceleration time |

The items of FS15*i* cover FS15. For a parameter number of FS15 different from FS15*i*, the item is described separately.
FS18*i*, 20*i*, 21*i*, PM*i*-D, 18, 20, and 21 are covered by the items of FS16*i*/16.

## 3.4.1 α Series and α (HV) Series Spindle Amplifier Module

If an alarm occurs in the spindle amplifier module, the ALM LED lights red in the STATUS display, and the two-digit 7-segment LEDs indicate the alarm code.




### 3.4.1.1 Alarm codes A, A0 to A4, and other Ax (x for representing an arbitrary number)

A, A0 : The SPM control program is not operating.
A1, A2, Ax : An error was detected in SPM control program processing.
A3 : An error was detected in the clock of the SPM control printed circuit board.
A4 : A parity error was detected in the SRAM on the SPM control printed circuit board.

(1) If an alarm is issued when the power to the PSM is turned on
  (a) The ROM is not installed correctly, no ROM is installed, or the ROM specification is incorrect
  If the ROM on the SPM control printed circuit board is once replaced, check the following:
  <1> Installation direction
  <2> Occurrence of a bent lead and so forth
  <3> Matching of the ROM series with the unit. Check Subsection 2.2.3 of Part I.
  (b) Printed circuit board failure
  Replace the SPM or SPM control printed circuit board.

(2) If an alarm is issued during motor activation
  (a) Influence of noise
  Referring to Chapter 5, "Installation," of "FANUC SERVO AMPLIFIER α series Descriptions (B-65162E)," check the GND-related wiring. If the signal cable of the spindle sensor is bundled with the power line of the servo motor, separate them from each other.

### 3.4.1.2 Alarm code 01

The inside temperature of the motor is higher than the specified temperature.

(1) If this alarm is issued during cutting (the motor temperature is high)
  (a) Check the cooling state of the motor.
    <1> If the cooling fan of the spindle motor is stopped, check the power supply of the cooling fan. If the cooling fan is still inoperative, replace it with a new one.
    <2> When a liquid-cooled motor is used, check the cooling system.
    <3> When the ambient temperature of the spindle motor is higher than the specified temperature, lower the ambient temperature to satisfy the specification.
  (b) If this alarm is issued even when the load meter fluctuates in a limited range, check the short-period rating. If the specified value is exceeded, reduce the load.

(2) If this alarm is issued under a light load (the motor temperature is high)
  (a) When the frequency of acceleration/deceleration is too high
  Set such a use condition that the average including output at acceleration/deceleration does not exceed the continuous rating.
  (b) The parameters specific to the motor are not set correctly.
  Refer to "FANUC AC SPINDLE MOTOR α series Parameter Manual (B-65160E)."

(3) If this alarm is issued when the motor temperature is low
  (a) The spindle sensor feedback cable is faulty.
  Replace the cable.
  (b) The SPM control printed circuit board is faulty.
  Replace the SPM control printed circuit board or SPM.
  (c) The motor (internal thermostat) is faulty.
  Replace the motor.

## 3.4.1.3 Alarm code 02

The actual motor speed is largely deviated from the commanded speed.

(1) If this alarm is issued during motor acceleration

(a) The parameter setting of acceleration/deceleration time is incorrect.
In the following parameter, set the value equivalent to the spindle inertia plus some margin:

| FS0 | FS15*i* | FS16*i*/16, PM-D/F | Description |
|---|---|---|---|
| 6582 | 3082 | 4082 | Setting of acceleration/deceleration time |

(b) The parameter for the speed detector is not set correctly.
Referring to "FANUC AC SPINDLE MOTOR α series Parameter Manual (B-65160E)," set a correct value.

(2) If this alarm is issued during cutting

(a) The cutting load has exceeded the motor output power.
Check the load meter indication, and review the use condition.

(b) The parameters for output restriction are not set correctly.
Check that the settings of the following parameters satisfy the machine and motor specifications:

| FS0 | FS15*i* | FS16*i*/16, PM-D/F | Description |
|---|---|---|---|
| 6528 | 3028 | 4028 | Output restriction pattern setting |
| 6529 | 3029 | 4029 | Output restriction value |

(c) The parameters specific to the motor are not correctly.
Refer to "FANUC AC SPINDLE MOTOR α series Parameter Manual (B-65160E)."

### 3.4.1.4 Alarm code 03

The fuse of the DC link has blown. (The voltage at the DC link is insufficient.) This alarm is checked when emergency stop is cancelled.

(1) If this alarm is issued during spindle operation (rotation)
The fuse of the DC link inside the SPM has probably blown. So, replace the SPM. This alarm may be caused by the following:
<1> Power line short-circuited to ground
<2> Motor winding short-circuited to ground
<3> IGBT or IPM module failure

(2) If the PSM input magnetic contactor is once turned on and is turned off with this alarm when emergency stop is cancelled or the CNC is started (When two spindles are connected, the magnetic contactor may not be turned off.)
(a) The DC link wire is not connected.
Check the DC link (TB1) wiring for errors.
(b) A cable is faulty.
Pin 9 of the interface cable (JX1B-JX1A) between the PSM and SPM may be short-circuited to 0V. Replace the cable.
(c) The fuse of the DC link inside the SPM has blown.
Replace the SPM.

### 3.4.1.5 Alarm code 04

It is detected that the 3-phase input to the main circuit of the PSM has an open phase.

(1) If the PSM indicates alarm code 6
Troubleshoot according to the description of alarm code 6 provided in Section 3.1 of Part II.

(2) If the PSM indicates no alarm
(a) A cable is faulty.
The connection cable between the PSM (JX1B) and SPM (JX1A) is faulty. Replace the cable.
(b) The SPM is faulty.
Replace the SPM or SPM control printed circuit board.

## 3.4.1.6 Alarm code 07

The motor rotates at a speed exceeding 115% (standard setting) of the maximum allowable speed.

(1) If this alarm is issued during spindle synchronization
If the activation (SFR, SRV) of the motor on one side is turned off then back on in the spindle synchronization mode, the spindle motor is accelerated to correct a position error built up during the period, thus causing this alarm. Modify the ladder so that SFR and SRV are not turned off in the spindle synchronization mode.

(2) If this alarm is issued while the motor is stopped
(a) The connection cable of the spindle sensor is faulty.
Check if the cable of the spindle sensor built into the motor (BZ sensor when a built-in motor is used) is disconnected, and replace the cable as required.
(b) The spindle sensor built into the motor is not adjusted correctly.
Adjust the sensor according to Subsection 4.3.4 of Part I.

## 3.4.1.7 Alarm code 09

The temperature of the heat sink of the SPM main circuit has risen to the set value. This alarm is issued for SPM-15 and up or SPM-15HV and later. With SPM-2.2 to SPM-11 and SPM-11HV, however, alarm code 12 is issued for the same cause.

(1) If this alarm is issued during cutting (the heat sink temperature is high)
(a) If this alarm is issued when the load meter reads a value below the continuous rating of the amplifier, check the cooling state of the heat sink.
<1> If the cooling fan is stopped, check the power supply (connector CX1A/B). If the cooling fan is still inoperative, replace the SPM with a new one.
<2> When the ambient temperature is higher than the specified temperature, lower the ambient temperature to satisfy the specification.
(b) When this alarm is issued because the load meter reads a value above the continuous rating of the amplifier, improve the use method.